

A PROJECT REPORT
ON

Ai-Powered Chatbot Enquiry System

Submitted in the fulfillment of the requirements for the
Award of the degree
of

B. TECH COMPUTER SCIENCE AND DESIGN

T.E. Semester VI

Engineering Design and Innovation IV

SUBMITTED BY

- | | |
|---------------------|--------------------|
| 1) Manthan Ahire | Exam No: BTCSD6003 |
| 2) Harshal Bonde | Exam No: BTCSD6011 |
| 3) Tejas Holkar | Exam No: BTCSD6015 |
| 4) Nikhil Malichkar | Exam No: BTCSD6025 |

Under the Guidance of
Priyanka Pawar



DR. D.Y. PATIL SCHOOL OF SCIENCE AND TECHNOLOGY, TATHAWADE,
PUNE.

2025 – 2026



CERTIFICATE

This is to certify that the Project report entitled

“ Ai-Powered Chatbot Enquiry System ”

submitted by

1) Manthan Ahire

Exam No: BTCSD6003

2) Harshal Bonde

Exam No: BTCSD6011

3) Tejas Holkar

Exam No: BTCSD6015

4) Nikhil Malichkar

Exam No: BTCSD6025

Is a Bonafide student of this institute and the work has been carried out by him/her under the supervision of Ms. Priyanka Pawar and it is approved for partial fulfillment for the requirement of Dr. D.Y. Patil Vidyapeeth for the award of the degree B.TECH in Computer Science and Design (T.E. Semester VI, Engineering Design and Innovation IV).



Project Guide

Head of the Department

ACKNOWLEDGEMENT

It is our privilege to express our sincerest regards to our project guide, Ms. Sayli Haldavanekar & Ms. Priyanka Pawar, for her valuable inputs, able guidance, encouragement, whole-hearted cooperation and constructive criticism throughout the duration of our project. We deeply express our sincere thanks to our Head of Department. Dr. Sonali Patil for encouraging and allowing us to present the project on the topic “AI powered Chatbot” at our department premises. We take this opportunity to thank all our faculties who directly or indirectly helped our project. We pay our respect and love to our parents and all other family members and friends for their love and encouragement throughout our career. Last but not the least we express our thanks to our friends for their cooperation and support.

ABSTRACT

In modern educational institutions, accessing accurate and timely information is essential for both students and visitors. However, traditional methods—such as manually asking administrative staff or browsing through static web pages—often fail to provide real-time, interactive, and user-friendly responses. To address this issue, this project presents an AI-powered chatbot system designed to assist users in quickly obtaining essential college-related information through a website interface. The chatbot leverages Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to understand user queries and respond conversationally. It is integrated with a structured database that stores key institutional information, including details about the Head of Department (HOD), college director, faculty members, class schedules, and student feedback. By simulating human-like conversation, the chatbot delivers accurate, context-aware responses, ensuring a smooth and intuitive user experience. This system aims to enhance accessibility and efficiency by allowing users to retrieve information in real time without depending on manual assistance. Additionally, it reduces the workload on administrative staff by automating repetitive queries. Overall, the AI chatbot acts as a digital assistant that simplifies communication, improves the user experience, and supports the institution's move toward smart and interactive education systems.

Table of Contents

Topic	Page No.
Acknowledgement	
Abstract	
1. INTRODUCTION	
1.1. Overview	
1.2. Motivation	
1.3. Problem Statement	
2. LITERATURE SURVEY	
3. REQUIREMENT SPECIFICATIONS	
3.1. Software Requirements	
3.2. Functional Requirements	
3.3. Non-Functional Requirements	
4. SYSTEM DESIGN	
4.1. Use case Diagram	
4.2. Activity Diagram	
4.3. System Architecture	
5. IMPLEMENTATION DETAILS	
5.1. Description of Tools	
5.2. Programming Language Description	
5.3. Output of Implementation (In form of Screen Shots)	

INTRODUCTION

1.1 Overview

The AI-Powered College Information Chatbot is a web-based application designed to make accessing college-related information faster, easier, and more interactive. It serves as a virtual assistant that allows students, faculty, and visitors to get answers to their queries in real time through a conversational interface. The chatbot eliminates the need to search through multiple web pages or contact staff for basic information, saving both time and effort. This system integrates Artificial Intelligence (AI) and Natural Language Processing (NLP) to understand user questions and provide accurate responses from a structured college database. The database stores key details such as information about the college director, Head of Department (HOD), faculty members, class timetables, and student reviews. When users interact with the chatbot, it processes their input, matches it with the relevant data, and displays the required information instantly. The chatbot is built to be user-friendly, responsive, and efficient, ensuring smooth interaction and accessibility across various devices. It helps reduce the workload on administrative and teaching staff by automating frequently asked queries, while also providing 24/7 availability for users. Overall, this project represents a step toward digital transformation in education, offering an intelligent and interactive communication platform that enhances user experience, promotes automation, and supports the vision of a smart and connected campus.

1.2 Motivation

The motivation behind developing the modern era of digital transformation, educational institutions are expected to provide fast, efficient, and reliable access to information. However, many colleges still depend on traditional methods such as notice boards, printed schedules, and manual inquiries for sharing updates and essential information. These methods are often timeconsuming and inconvenient, especially when students or visitors need quick answers. The motivation behind developing an AI-Powered College Information Chatbot arises from the need to overcome these communication gaps and make information access more interactive and user-friendly. With advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP), it is now possible to create systems that can understand human language and respond intelligently. This inspired the idea of designing a chatbot that could act as a virtual assistant, providing instant responses to common queries related to the college. By automating routine inquiries—such as details about the Head of Department, faculty members, timetables, and student feedback—the chatbot reduces the workload on administrative staff and enhances the overall efficiency of the institution. Moreover, it allows students and visitors to get realtime information anytime and from anywhere, making the system more accessible and convenient. This project is driven by the vision of creating a smart educational environment where technology bridges the communication gap between students, faculty, and administration. The goal is to improve user experience, promote automation, and contribute to the digital modernization of educational institutions.

1.3 Problem Statement

In most educational institutions, students and visitors often face difficulties in accessing accurate and timely information related to faculty details, class schedules, departmental contacts, and other college-related queries. The traditional methods—such as checking notice boards, visiting administrative offices, or browsing multiple web pages—are time-consuming, inefficient, and often fail to provide real-time updates. Moreover, staff members spend a significant amount of time addressing repetitive questions that could easily be automated. This creates unnecessary workload, delays communication, and affects overall institutional efficiency. Therefore, there is a need for an intelligent, automated system that can provide instant, accurate, and interactive responses to user queries. The proposed AI-Powered College Information Chatbot aims to solve this problem by using Artificial Intelligence (AI) and Natural Language interact naturally and obtain relevant college information quickly and efficiently.

LITERATURE SURVEY

The development of conversational agents has evolved significantly over the past decade. Early research in end-to-end conversational modeling demonstrated that neural networks could generate natural-sounding dialogue responses without relying on rigid rule-based structures. Vinyals and Le (2015) introduced a neural conversational model that could predict the next utterance based on previous dialogue context, marking a major step forward in dialogue automation (arXiv). Later, the introduction of the Transformer architecture revolutionized Natural Language Processing (NLP) by replacing recurrent structures with a scalable selfattention mechanism. This advancement allowed for better context handling across sentences and improved language understanding, making transformers the dominant framework for modern conversational systems (arXiv). Additionally, pre-trained language models like BERT further enhanced NLP by providing deep bidirectional contextual understanding that could be fine-tuned for specific tasks such as intent recognition, question answering, and named-entity detection — all crucial elements in chatbot design (ACL Anthology). While purely neural models improved language fluency, they often lacked factual accuracy. To address this, retrieval-augmented generation (RAG) techniques emerged, combining neural retrievers with text generators to ground responses in verified external knowledge sources (NeurIPS Proceedings). This approach is particularly effective for knowledge-intensive tasks such as delivering up-to-date college information, where accuracy and reliability are critical. By integrating structured databases and document retrieval, RAG-based systems can deliver both intelligent and trustworthy answers to user queries. In the field of education, numerous studies have shown the benefits of chatbot integration. Research published in ScienceDirect and similar platforms highlights improvements in accessibility, personalized learning support, and staff efficiency when chatbots are deployed in educational settings. Chatbots can provide immediate responses to students' academic and administrative questions, increasing engagement and satisfaction levels. However, studies also point out challenges related to system maintenance, privacy, and data integration. Case studies from large universities (Axios) demonstrate how chatbots significantly reduce staff workload, improve response times, and enhance the overall user experience when properly designed and maintained. The literature also identifies common architectural components used in educational chatbots. These typically include Natural Language Understanding (NLU) modules for intent detection and entity extraction, Dialogue Managers for maintaining multiturn context, and a Knowledge Layer consisting of structured databases or document indexes for storing relevant information. The Response Generation module may use either template-based or retrieval-augmented generative methods, balancing accuracy with flexibility. Finally, integration components such as chat widgets, authentication layers, and fallback systems connect the chatbot to users through institutional websites. Each architectural decision involves trade-offs: template-based systems provide higher accuracy but limited flexibility, while generative models offer natural conversation flow but risk inaccuracies without proper grounding. Evaluation methods discussed across studies include both automatic and human-centric approaches. Automatic metrics such as intent classification accuracy, slot filling F1 scores, and BLEU/ROUGE scores help measure linguistic and functional performance. Human evaluations assess user satisfaction, helpfulness, and efficiency, while operational measures like reduction

in staff queries and system uptime gauge real-world effectiveness. Literature suggests that a combination of quantitative and qualitative evaluations provides the most comprehensive assessment for educational chatbots. Despite their potential, several challenges persist. Maintaining data freshness and provenance is vital since institutional information changes frequently. Privacy and access control are major concerns, especially when handling student or administrative data. Generative models also carry the risk of hallucination—producing incorrect or fabricated information—if not properly constrained. Moreover, ensuring smooth user experiences in multi-turn dialogues and integrating chatbots with existing databases require careful engineering and governance. A clear gap identified in the literature is that while many chatbots demonstrate educational potential, few offer a fully deployable system that combines structured databases with modern NLP techniques and retrieval-based accuracy. Most existing systems are research prototypes or limited to narrow academic assistance.

REQUIREMENT SPECIFICATIONS

3.1 Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Python 3.8 or higher
- Framework: Flask (for web server and routing)
- Database: SQLite (for storing book records and metadata)
- Frontend: HTML, CSS (for the chat interface)
- Browser: Google Chrome, Mozilla Firefox, or Microsoft Edge
- Code Editor: Visual Studio Code, PyCharm
- Additional Libraries: spacy (for basic Natural Language Processing)

3.2 Functional Requirements

- User Interaction: Accept user queries in natural language format.
- Intent Recognition: Detect and classify user intents (e.g., search, suggest, availability check).
- Database Query Execution: Retrieve accurate results from the SQLite database.
- Response Generation: Display chatbot responses in a conversational chat interface.
- Error Handling: Gracefully manage unclear or invalid user inputs.
- Session Management: Maintain conversation context during a single user session.

3.4 Non-Functional Requirements

- Usability: Simple and intuitive UI for non-technical users.
- Reliability: Consistent operation with minimal downtime or crashes.
- Performance: Response time should be less than 3 seconds per query.
- Scalability: System should allow easy addition of new books or categories.
- Security: Restrict unauthorized database access and input injection.
- Maintainability: Code should be modular and well-documented for easy updates.
- Portability: Should run smoothly across Windows, Linux, and macOS systems.
- Extensibility: Support integration of voice commands or cloud-based databases in the future

PERIODICALS

- [1] J. F. Fuller, E. F. Fuchs, and K. J. Roesler, "Influence of harmonics on power distribution system protection," *IEEE Trans. Power Delivery*, vol. 3, pp. 549-557, Apr. 1988.

Books:

- [1] E. Clarke, *Circuit Analysis of AC Power Systems*, vol. I. New York: Wiley, 1950, p. 81.
- [2] G. O. Young, "Synthetic structure of industrial plastics," in *Plastics*, 2nd ed., vol. 3, J. Peters, Ed. New York: McGraw-Hill, 1964, pp. 15-64.

Technical Reports:

- [1] E. E. Reber, R. L. Mitchell, and C. J. Carter, "Oxygen absorption in the Earth's atmosphere," Aerospace Corp., Los Angeles, CA, Tech. Rep. TR-0200 (4230-46)3, Nov. 1968.
- [2] S. L. Talleen. (1996, Apr.). The Intranet Architecture: Managing information in the new paradigm. Amdahl Corp., Sunnyvale, CA. [Online].
Available: <http://www.amdahl.com/doc/products/bsg/intra/infra/html>

Papers Presented at Conferences (Unpublished):

- [1] D. Ebehard and E. Voges, "Digital single sideband detection for interferometric sensors," presented at the 2nd Int. Conf. Optical Fiber Sensors, Stuttgart, Germany, 1984.
- [2] Process Corp., Framingham, MA. Intranets: Internet technologies deployed behind the firewall for corporate productivity. Presented at INET96 Annu. Meeting. [Online]. Available: <http://home.process.com/Intranets/wp2.htm>

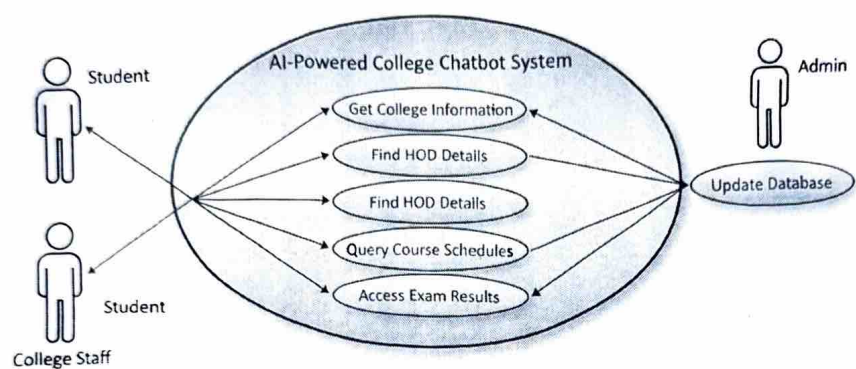
Papers from Conference Proceedings (Published):

- [1] J. L. Alquieres and J. C. Praca, "The Brazilian power system and the challenge of the Amazon transmission," in Proc. 1991 IEEE Power Engineering Society Transmission and Distribution Conf., pp.315-320.

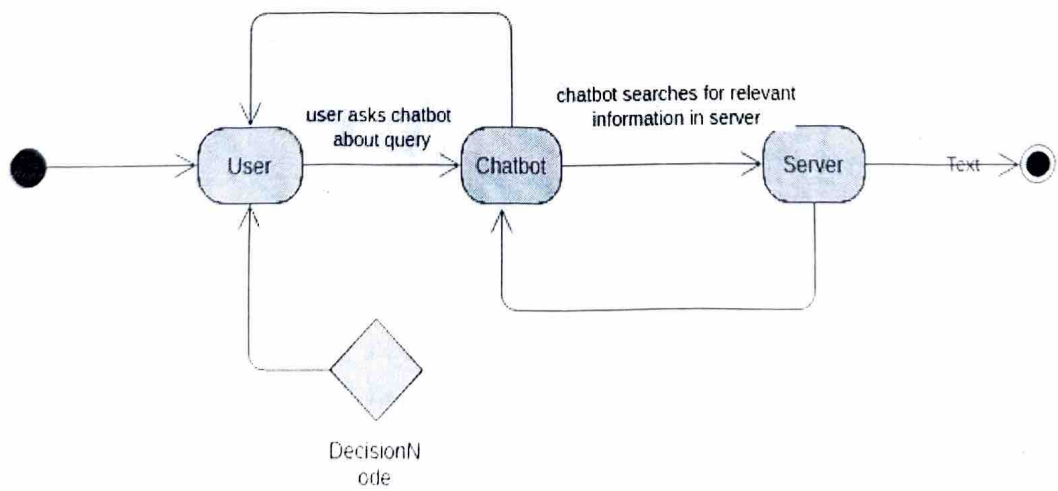
SYSTEM DESIGN

4.1 Use Case Diagram

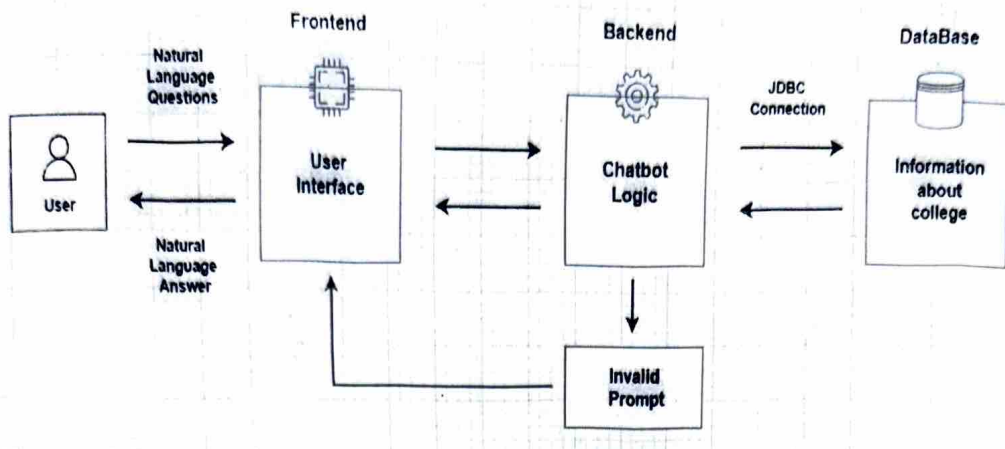
Use Case Diagram: AI College Chatbot



4.2 Activity Diagram



4.3 System Architecture



IMPLEMENTATION DETAILS

5.1 Description of Tools

- **Python:** Used for backend development, implementing logic, and handling data manipulation.
- **Flask Framework:** Provides lightweight web server and RESTful API routing to connect frontend and backend.
- **SQLite Database:** Chosen for simplicity, minimal configuration, and compatibility with
- **HTML, CSS, JavaScript:** Used to design and style the user-friendly chat interface.
- **NLTK (Optional):** Utilized for preprocessing (tokenization, stop-word removal) and basic intent recognition.
- **Jinja2 Templates:** Used in Flask to dynamically inject responses into HTML.
- **Bootstrap (Optional):** Used to enhance the interface design for responsiveness

5.2 Programming Language Description

Python's readability, extensive library support (Flask, SQLite3, NLTK), and ease of integration made it the ideal language for implementing the chatbot. Flask was chosen over heavier frameworks like Django for its simplicity and minimal setup, enabling faster testing and deployment. SQLite was used for its lightweight and serverless nature, which fits small to medium-sized applications like this one.

5.3 Output of Implementation

The first screenshot shows the login page of the DYPSSST Enquiry Bot. The header includes 'DYPSSST Enquiry Bot' and 'Login Register'. The main heading is 'Dr. D. Y. Patil School of Science and Technology' with the subtitle 'Student enquiry support portal'. The login form has fields for 'Email' (filled with 'bajeshkar553@gmail.com') and 'Password' (masked with dots). Below the fields is a 'Login to Dashboard' button. At the bottom of the form are links for 'Create account' and 'Forgot password?'.

The second screenshot is identical to the first, showing the login page.

The third screenshot shows the chat interface after a successful login. The header includes 'DYPSSST ENQUIRY BOT' and navigation links 'Home', 'About Us', 'Suggestion box', and 'Log Out'. A success message 'You were successfully logged in' is displayed. The chat area shows two messages from the 'DYPSSST BOT': 'Hi there, welcome to DYPSSST! If you need assistance, I am always here.' and 'Which of the following user groups do you belong to?'. Below these are four numbered options: '1. Student's Section Enquiry.', '2. Faculty Section Enquiry.', '3. Parent's Section Enquiry.', and '4. Visitor's Section Enquiry.'. At the bottom, there is an input field 'Enter your enquiry number...' and a 'Send' button.

CONCLUSION AND RESULTS

6.1 Conclusion

In summary, this AI-powered assistant chatbot represents a modern and innovative solution designed to enhance communication between students, parents, and the college administration team. By leveraging advanced Natural Language Processing (NLP) techniques and integrating with a centralized database, the chatbot is capable of providing instant, accurate, and userfriendly responses to a wide variety of queries. This not only saves time for students and parents seeking information but also significantly reduces the workload and repetitive tasks for college staff, allowing them to focus on more critical and strategic responsibilities. The chatbot ensures that information is easily accessible at any time, offering 24/7 assistance that a traditional helpdesk or static website cannot match. Unlike conventional college websites that often provide fixed and limited information, this intelligent chatbot delivers dynamic, personalized, and context-aware responses tailored to each user's needs. Whether it's answering questions about admissions, course details, exam schedules, fees, or campus facilities, the chatbot acts as a reliable digital guide for all stakeholders.

6.2 Results

The **output** layer serves as the stage where the chatbot delivers its responses to the user in a clear, **engaging**, and user-friendly manner. Depending on the nature of the query, the format of the response can differ. For straightforward questions, the chatbot provides concise text-based answers, such as "Dr. Sonali Patil is the Head of the Computer Science Department." For more **structured** data, it may present information in a tabular format, like class schedules or **timetables**, displayed in a neatly organized table. When summarizing feedback or listing multiple names, the chatbot uses bullet points or ordered lists for better readability. In addition, it can **generate** interactive responses, prompting follow-up options such as "Would you like to view the Director's details as well?" This layer plays a crucial role in ensuring that all outputs are well-formatted, easy to comprehend, and contextually relevant.